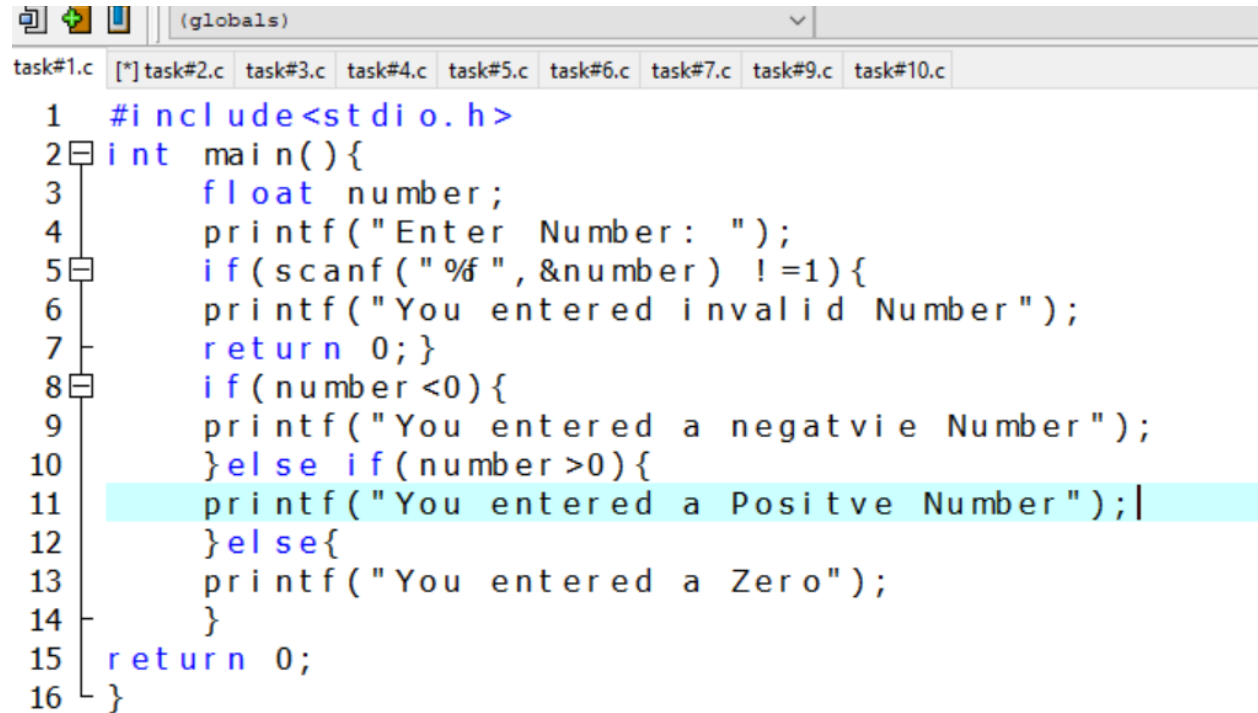


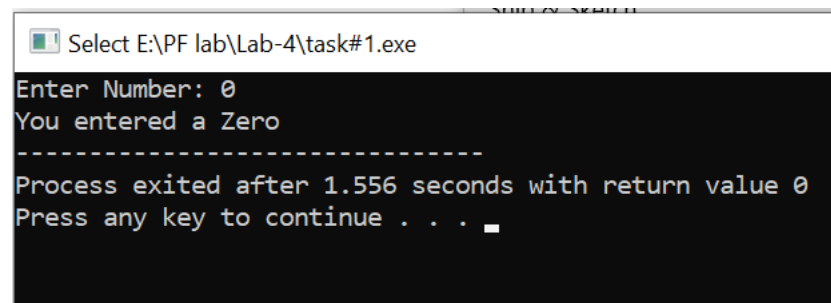
Programing Fundamental Lab#4

Task#1

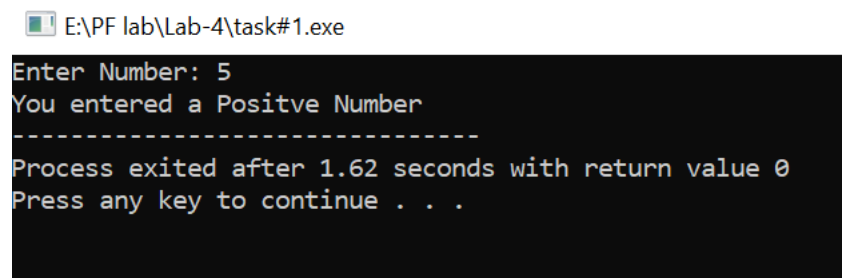


```
task#1.c  (*) task#2.c  task#3.c  task#4.c  task#5.c  task#6.c  task#7.c  task#9.c  task#10.c
1  #include<stdio.h>
2  int main(){
3      float number;
4      printf("Enter Number: ");
5      if(scanf("%f", &number) !=1){
6          printf("You entered invalid Number");
7          return 0; }
8      if(number<0){
9          printf("You entered a negatvie Number");
10     }else if(number>0){
11         printf("You entered a Positive Number");
12     }else{
13         printf("You entered a Zero");
14     }
15     return 0;
16 }
```

Results



```
Select E:\PF lab\Lab-4\task#1.exe
Enter Number: 0
You entered a Zero
-----
Process exited after 1.556 seconds with return value 0
Press any key to continue . . .
```



```
E:\PF lab\Lab-4\task#1.exe
Enter Number: 5
You entered a Positive Number
-----
Process exited after 1.62 seconds with return value 0
Press any key to continue . . .
```

Programing Fundamental Lab#4

E:\PF lab\Lab-4\task#1.exe

```
Enter Number: -6
You entered a negativie Number
-----
Process exited after 1.607 seconds with return value 0
Press any key to continue . . .
```

Task#2

```
task#1.c task#2.c task#3.c task#4.c task#5.c task#6.c task#7.c task#9.c task#10.c
1 #include <stdio.h>
2 int main() {
3     int NumberOne, NumberTwo;
4     printf("-----X-----\n");
5     printf("Simple Calculator\n");
6     printf("-----X-----\n");
7     printf("Enter a Number: ");
8     scanf("%d", &NumberOne);
9     printf("Enter Another Number: ");
10    scanf("%d", &NumberTwo);
11    int number;
12    printf("For Addition enter 1\nFor Subtraction enter 2 \nFor Multiplication enter 3 \nFor division enter 4\n");
13    printf("Enter Number of Operation: ");
14    scanf("%d", &number);
15    switch (number)
16    {
17    case 1:
18        printf("The Sum of Number One and Number two= %d \n", NumberOne+NumberTwo);
19        break;
20    case 2:
21        printf("The Difference of Number One and Number two= %d \n", NumberOne-NumberTwo);
22        break;
23    case 3:
24        printf("The Product of Number One and Number two= %d \n", NumberOne*NumberTwo);
25        break;
26    case 4:
27        if (NumberTwo == 0) {
28            printf("Division by Zero is not possible\n");
29        } else {
30            printf("The Division of Number One and Number two= %d \n", NumberOne / NumberTwo);
31        }
32        break;
33    default:
34        printf("You entered invalid Number");
35        break;
36    }
37    return 0;
38 }
39
40 }
```

Results

```
Enter a Number: 5
Enter Another Number: 5

Choose your operation:
For Addition press 1
For Subtraction press 2
For Multiplication press 3
For Division press 4
Enter Number of Operation: 4
The Division of Number One by Number Two is: 1
-----
Process exited after 4.755 seconds with return value 0
Press any key to continue . . .
```

Programing Fundamental Lab#4

```
E:\PF lab\Lab-4\task#2.exe
Enter a Number: 5
Enter Another Number: 5
Choose your operation:
For Addition press 1
For Subtraction press 2
For Multiplication press 3
For Division press 4
Enter Number of Operation: 1
The Sum of Number One and Number Two is: 10
-----
Process exited after 3.667 seconds with return value 0
Press any key to continue . . .
```

```
E:\PF lab\Lab-4\task#2.exe
Enter a Number: 7
Enter Another Number: 4
Choose your operation:
For Addition press 1
For Subtraction press 2
For Multiplication press 3
For Division press 4
Enter Number of Operation: 2
The Difference between Number One and Number Two is: 3
-----
```

```
E:\PF lab\Lab-4\task#2.exe
Enter a Number: 3
Enter Another Number: 3
Choose your operation:
For Addition press 1
For Subtraction press 2
For Multiplication press 3
For Division press 4
Enter Number of Operation: 3
The Product of Number One and Number Two is: 9
-----
```

Programing Fundamental Lab#4

Task#3

```
task#1.c task#2.c task#3.c task#4.c task#5.c task#6.c task#7.c task#8.c task#9.c task#10.c [*] task2.c
1  #include<stdio.h>
2  int main () {
3      int WaterLevel;
4      printf("Enter the water level:");
5      scanf("%d", &WaterLevel);
6      if(WaterLevel > 6) {
7          printf("Evacuation Required");
8          return 0;
9      } else if(WaterLevel < 6 && WaterLevel > 4) {
10         printf("Medium Relief Package");
11         return 0;
12     } else if(WaterLevel < 4 && WaterLevel > 2) {
13         printf("Small Relief Package");
14         return 0;
15     } else {
16         printf("No Relief Required!");
17     }
18     return 0;
19 }
```

Results

```
E:\PF lab\Lab-4\task#3.exe
Enter the water level:5
Medium Relief Package
-----
Process exited after 2.056 seconds with return value 0
Press any key to continue . . .
```

Programing Fundamental Lab#4

```
Select E:\PF lab\Lab-4\task#3.exe
Enter the water level:3
Small Relief Package
-----
Process exited after 2.642 seconds with return value 0
Press any key to continue . . .
```

```
Enter the water level:7
Evacuation Required
-----
Process exited after 1.696 seconds with return value 0
Press any key to continue . . .
```

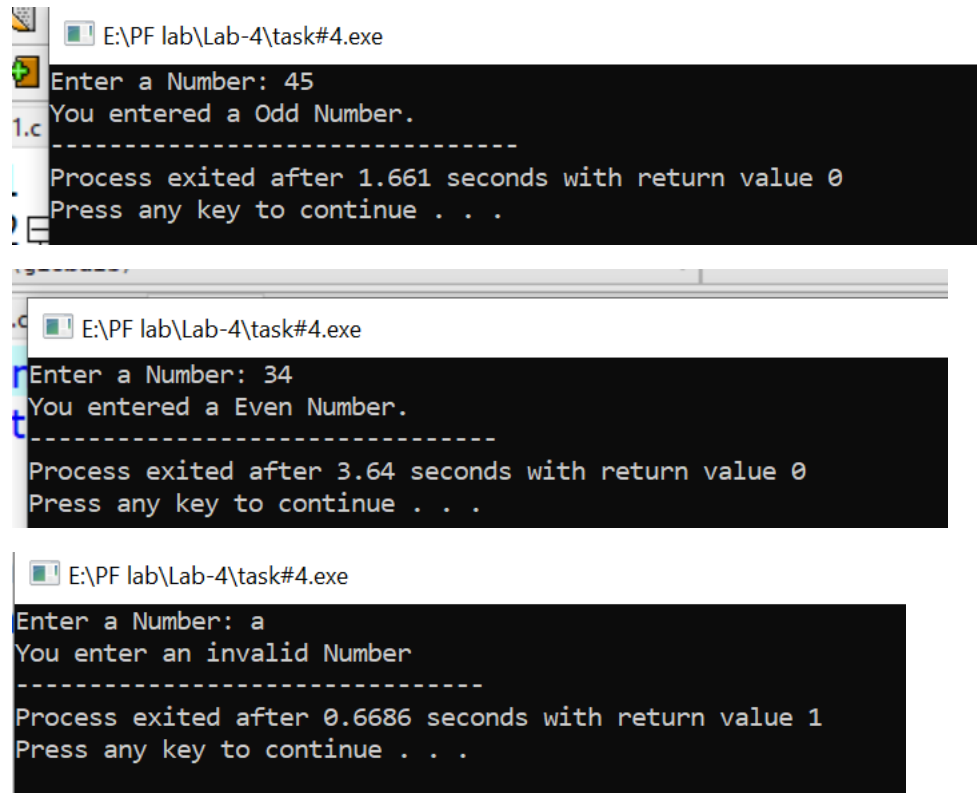
```
E:\PF lab\Lab-4\task#3.exe
Enter the water level:1
No Relief Required!
-----
Process exited after 2.11 seconds with return value 0
Press any key to continue . . .
```

Task#4

```
task#1.c task#2.c task#3.c task#4.c task#5.c task#6.c task#7.c task#8.c task#9.c task#10.c [*] task2.c
1 #include<stdio.h>
2 int main(){
3     int number;
4     printf("Enter a Number: ");
5     if(scanf("%d",&number)!=1){
6         printf("You enter an invalid Number");
7         return 1;
8     }
9     if(number%2==0){
10        printf("You entered a Even Number.");
11    }else{
12        printf("You entered a Odd Number.");
13    }
14    return 0;
15 }
16 }
```

Programing Fundamental Lab#4

Results



The image displays three sequential screenshots of a Windows command prompt window running an executable file named 'E:\PF lab\Lab-4\task#4.exe'. Each screenshot shows the program's output for a specific user input.

First Screenshot: The user enters '45'. The program outputs 'Enter a Number: 45', 'You entered a Odd Number.', a separator line of dashes, 'Process exited after 1.661 seconds with return value 0', and 'Press any key to continue . . .'. The window title is 'E:\PF lab\Lab-4\task#4.exe'.

Second Screenshot: The user enters '34'. The program outputs 'Enter a Number: 34', 'You entered a Even Number.', a separator line of dashes, 'Process exited after 3.64 seconds with return value 0', and 'Press any key to continue . . .'. The window title is 'E:\PF lab\Lab-4\task#4.exe'.

Third Screenshot: The user enters 'a'. The program outputs 'Enter a Number: a', 'You enter an invalid Number', a separator line of dashes, 'Process exited after 0.6686 seconds with return value 1', and 'Press any key to continue . . .'. The window title is 'E:\PF lab\Lab-4\task#4.exe'.

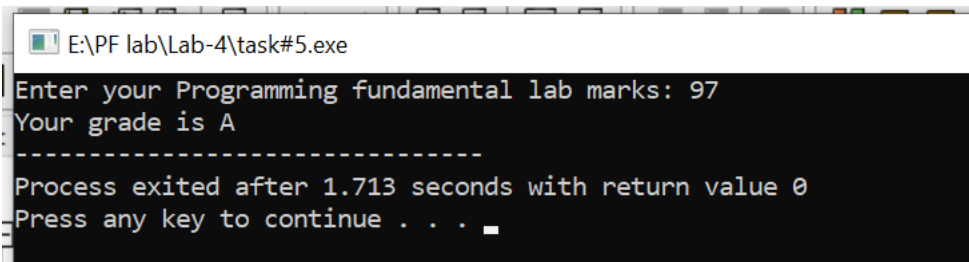
Programing Fundamental Lab#4

Task#5

```
task#2.C task#3.C task#4.C task#5.C task#6.C task#7.C task#8.C task#9.C task#10.C task#11.C task#12.C
# C:\Program Files (x86)\Dev-Cpp\MinGW64\x86_64-mingw32\include\stdio.h - Ctrl+ Click for more info
int main(){
    float Marks;
    printf("Enter your Programming fundamental lab marks: ");
    scanf("%f", &Marks);
    if(Marks >=85 && Marks <=100){
        printf("Your grade is A");

    }else if(Marks <=84 && Marks >=75){
        printf("Your grade is B");
    }else if(Marks <=69 && Marks >=55 ){
        printf("Your grade is c");
    }
    else if(Marks <=54 && Marks >=40){
        printf("Your grade is D");
    }else if(Marks <40&& Marks >=0){
        printf("Your grade is f");
    }else{
        printf("Invalid Marks");
    }
    return 0;
}
```

Result



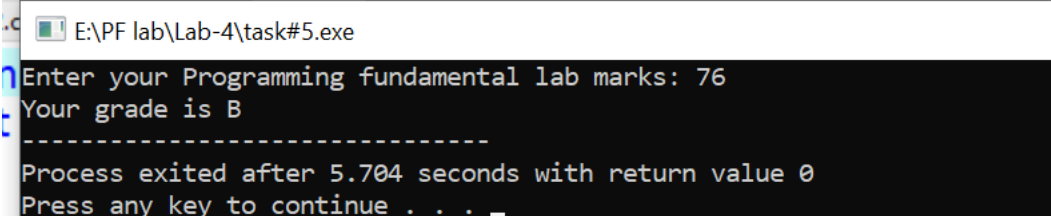
E:\PF lab\Lab-4\task#5.exe

Enter your Programming fundamental lab marks: 97

Your grade is A

Process exited after 1.713 seconds with return value 0

Press any key to continue . . .



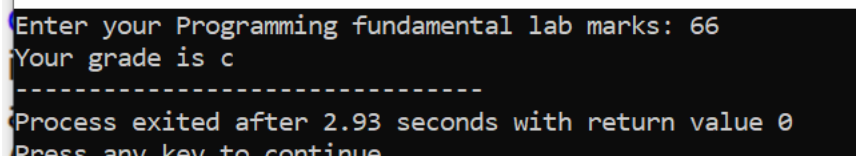
E:\PF lab\Lab-4\task#5.exe

Enter your Programming fundamental lab marks: 76

Your grade is B

Process exited after 5.704 seconds with return value 0

Press any key to continue . . .



Enter your Programming fundamental lab marks: 66

Your grade is c

Process exited after 2.93 seconds with return value 0

Press any key to continue . . .

Programing Fundamental Lab#4

E:\PF lab\Lab-4\task#5.exe

```
Enter your Programming fundamental lab marks: 54
Your grade is D
-----
Process exited after 4.045 seconds with return value 0
Press any key to continue . . .
```

E:\PF lab\Lab-4\task#5.exe

```
Enter your Programming fundamental lab marks: 39
Your grade is f
-----
Process exited after 3.17 seconds with return value 0
Press any key to continue . . .
```

Task#6

```
task#1.c task#2.c task#3.c task#4.c task#5.c [*] task#6.c task#7.c task#8.c task#9.c task#10.c [*] task2.c
1  #include<stdio.h>
2  int main(){
3      int a, b, c;
4      printf("Enter the Value of a: ");
5      if(scanf("%d", &a) != 1){
6          printf("Invalid Value of A\n");
7          return 0;
8      }
9      printf("Enter the Value of b: ");
10     if(scanf("%d", &b) != 1){
11         printf("Invalid Value of B\n");
12         return 0;
13     }
14     printf("Enter the Value of c: ");
15     if(scanf("%d", &c) != 1){
16         printf("Invalid Value of C\n");
17         return 0;
18     }
19     if(a > b && a > c){
20         printf("The value of a is largest.\n");
21     } else if(b > a && b > c){
22         printf("The value of b is largest.\n");
23     } else if(c > a && c > b){
24         printf("The value of c is largest.\n");
25     } else if(a == b && a > c){
26         printf("The value of a and b are equal and largest.\n");
27     } else if(b == c && b > a){
28         printf("The value of b and c are equal and largest.\n");
29     } else if(a == c && a > b){
30         printf("The value of a and c are equal and largest.\n");
31     } else {
32         printf("All numbers have same value.\n");
33     }
34     return 0;
35 }
36
```


Programing Fundamental Lab#4

Results

```
E:\PF lab\Lab-4\task#6.exe
Enter the Value of a: 3
Enter the Value of b: 4
Enter the Value of c: 6
The value of c is largest.

-----
Process exited after 6.399 seconds with return value 0
Press any key to continue . . .
```

```
E:\PF lab\Lab-4\task#6.exe
Enter the Value of a: 65
Enter the Value of b: 5
Enter the Value of c: 4
The value of a is largest.

-----
Process exited after 4.286 seconds with return value 0
Press any key to continue . . .
```

```
E:\PF lab\Lab-4\task#6.exe
Enter the Value of a: 6
Enter the Value of b: 90
Enter the Value of c: 7
The value of b is largest.

-----
Process exited after 5.725 seconds with return value 0
Press any key to continue . . .
```

```
E:\PF lab\Lab-4\task#6.exe
Enter the Value of a: 6
Enter the Value of b: 6
Enter the Value of c: 6
All numbers have same value.

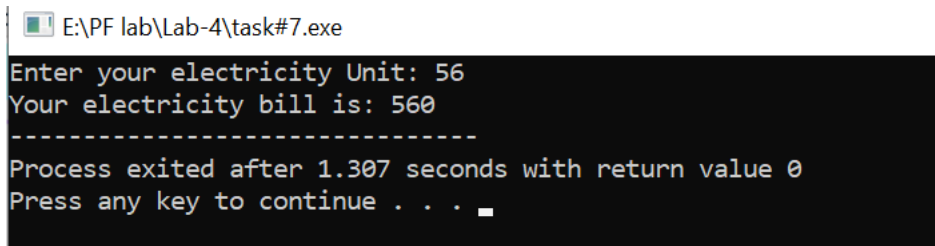
-----
Process exited after 2.592 seconds with return value 0
Press any key to continue . . .
```

Programing Fundamental Lab#4

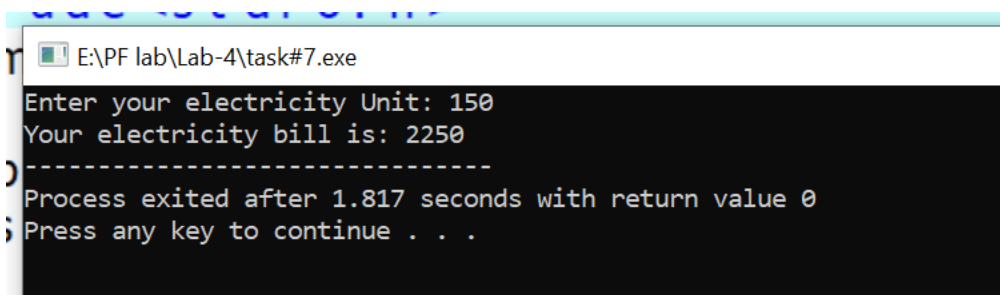
Task#7

```
#include<stdio.h>
int main(){
    int ElectricUnit;
    printf("Enter your electricity Unit: ");
    scanf("%d",&ElectricUnit);
    if(ElectricUnit<=100){
        printf("Your electricity bill is: %d",ElectricUnit*10);
    }else if(ElectricUnit>101 &&ElectricUnit<=300){
        printf("Your electricity bill is: %d",ElectricUnit*15);
    }else if(ElectricUnit>301 && ElectricUnit<=500){
        printf("Your electricity bill is: %d",ElectricUnit*20);
    }else if(ElectricUnit>=500){
        printf("Your electricity bill is: %d",ElectricUnit*25);
    }
    return 0;
}
```

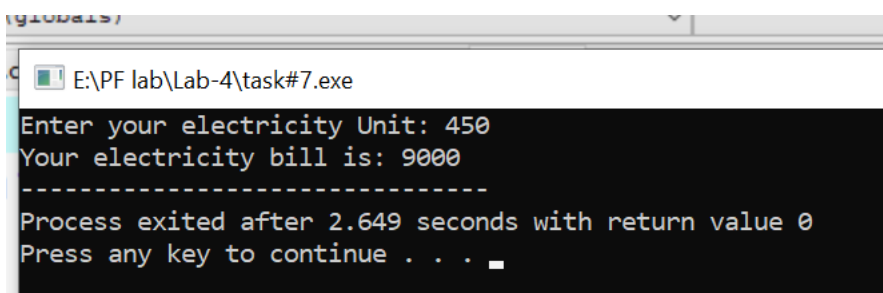
Results



```
E:\PF lab\Lab-4\task#7.exe
Enter your electricity Unit: 56
Your electricity bill is: 560
-----
Process exited after 1.307 seconds with return value 0
Press any key to continue . . .
```



```
E:\PF lab\Lab-4\task#7.exe
Enter your electricity Unit: 150
Your electricity bill is: 2250
-----
Process exited after 1.817 seconds with return value 0
Press any key to continue . . .
```



```
E:\PF lab\Lab-4\task#7.exe
Enter your electricity Unit: 450
Your electricity bill is: 9000
-----
Process exited after 2.649 seconds with return value 0
Press any key to continue . . .
```

Programing Fundamental Lab#4

ude <stdio.h>

E:\PF lab\Lab-4\task#7.exe

Enter your electricity Unit: 600

Your electricity bill is: 15000

Process exited after 1.094 seconds with return value 0

Press any key to continue . . .

Task#8

```
.c task#2.c task#3.c task#4.c task#5.c task#6.c task#7.c task#8.c task#9.c task#10.c [*] task2.c
#include <stdio.h>
#include <math.h>

int main() {
    int NumberOne, NumberTwo, operation;

    printf("Enter a Number: ");
    scanf("%d", &NumberOne);

    printf("Enter Another Number: ");
    scanf("%d", &NumberTwo);

    printf("\nChoose your operation:\n");
    printf("For Addition press 1\n");
    printf("For Subtraction press 2\n");
    printf("For Multiplication press 3\n");
    printf("For Division press 4\n");
    printf("For Modulus press 5\n");
    printf("For Power (NumberOne raised to NumberTwo) press 6\n");
    printf("Enter Number of Operation: ");
    scanf("%d", &operation);

    switch (operation) {
        case 1:
            printf("The Sum of Number One and Number Two is: %d\n", NumberOne + NumberTwo);
            break;
        case 2:
            printf("The Difference between Number One and Number Two is: %d\n", NumberOne - NumberTwo);
            break;
        case 3:
            printf("The Product of Number One and Number Two is: %d\n", NumberOne * NumberTwo);
            break;
        case 4:
            printf("The Division of Number One and Number Two is: %d\n", NumberOne / NumberTwo);
            break;
        case 5:
            printf("The Modulus of Number One and Number Two is: %d\n", NumberOne % NumberTwo);
            break;
        case 6:
            printf("Number One raised to the power of Number Two is: %2f\n", pow(NumberOne, NumberTwo));
            break;
        default:
            printf("You entered an invalid operation number.\n");
    }

    return 0;
}
```

Programing Fundamental Lab#4

Task#9

```
c task#2.c task#3.c task#4.c task#5.c task#6.c task#7.c [*]task#8.c task#9.c task#10.c [*]task2.c
#include<stdio.h>;
int main(){
    int RainFall=1445, RiverFlow=5450;

    if(RainFall<50 && RiverFlow<200){
        printf("Low Risk");
        return 0;
    }else if(RainFall>50 && RainFall<100 && RiverFlow>200 && RiverFlow<500){
        printf("Moderate Risk");
        return 0;
    }else if(RainFall>100 && RainFall<150 && RiverFlow>500 && RiverFlow<800){
        printf("High Risk");
        return 0;
    }else if(RainFall>150 && RiverFlow>800 ){
        printf("Severe Risk -Evacuate!");
    }
    return 0;
}
```

Task#10

```
k#1.c task#2.c task#3.c task#4.c task#5.c task#6.c task#7.c [*]task#8.c task#9.c task#10.c [*]task2.c
1 #include<stdio.h>
2 #include<stdlib.h>
3 int main(){
4     int AkhlaqManners, HonestyTrustworthiness, PrayerRegularity,
5     Fasting, ZakatCharity, SocialBehavior, ConflictSkills;
6
7     printf("-----x-----x-----\n");
8     printf("We are Calculating the Character of a man as per Islam\n");
9     printf("-----x-----x-----\n");
10
11     printf("Enter Akhlaq & Manners range 0-10: ");
12     scanf("%d", &AkhlaqManners);
13     // system(c);
14     printf("Enter Honesty &Trustworthiness range 0-10: ");
15     scanf("%d", &HonestyTrustworthiness);
16     printf("Enter Prayer Regularity range \n0 for Irregular,\n 1 for Regular: ");
17     scanf("%d", &PrayerRegularity);
18     printf("Enter Fasting range \n0 for Never,\n1 for Somtimes\n2 for Always : ");
19     scanf("%d", &Fasting);
20     printf("Enter Zakat & Charity range 0-10: ");
21     scanf("%d", &ZakatCharity);
22     printf("Enter Social Behavior range 0-10: ");
23     scanf("%d", &SocialBehavior);
24     printf("Enter Conflict Skills range 0-10: ");
25     scanf("%d", &ConflictSkills);
26     if (PrayerRegularity == 0) {
27         PrayerRegularity = 0;
28     } else {
29         PrayerRegularity = 10;
30     }
31 }
```

Programing Fundamental Lab#4

```

1 if (Fasting == 0) {
2     Fasting = 0;
3 } else if (Fasting == 1) {
4     Fasting = 5;
5 } else {
6     Fasting = 10;
7 }
8
9 system("cls");
10 printf("-----X-----\n");
11 printf("Character of a man as per Islam\n");
12 printf("-----X-----\n");
13 float CharacterIndex= (AkhlqaManners * 2.5)+(HonestyTrustworthiness*2.0)+(PrayerRegularity*1.0)+(Fasting*1.5)+(ZakatCharity*1.0)+(Soci
14
15 if (CharacterIndex>=85&&CharacterIndex<=100){
16     printf("Character: Excellent Muslim Character. \n");
17     printf("Remarks: Role Mode for society");
18     return 0;
19 } else if (CharacterIndex>=70&&CharacterIndex<85){
20     printf("Character: Good Muslim Character. \n");
21     printf("Remarks: Practicing Believer.");
22     return 0;
23 } else if (CharacterIndex>=50&&CharacterIndex<70){
24     printf("Character: Average Muslim Character. \n");
25     printf("Remarks: Needs minor improvement.");
26     return 0;
27 } else if (CharacterIndex>=30&&CharacterIndex<50){
28     printf("Character: Needs Improvement. \n");
29     printf("Remarks: Work on Akhlqa and Ibadat.");
30     return 0;
31 } else if (CharacterIndex<30){
32     printf("Character: Weak Character. \n");
33     printf("Remarks: Require serious guidance");
34     return 0;
35 } else{
36     printf("Invalid Input");
37 }
38
39 return 0;

```

Github link

<https://github.com/vishnani/Lab-4-task>